

Conjecture 00000002154 is false:
every cycle C_m is closed Eulerian, but its critical group maps onto
 $\mathbb{Z}/m$

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1 The conjecture

Conjecture 00000002154 (English text, verbatim): “Definition: The number of cyclic factors of the critical group $\text{Jac}(G)$ (the sandpile group) is the number of its cyclic $\mathbb{Z}$ -components. Conjecture: The classification of graphs with zero cyclic factors (i.e. Jac entirely cyclic-free): exactly subdivisions of closed Eulerian graphs (trivial exceptions exhausted).” The Chinese text says the same.

Reading. Graphs are finite simple graphs. For a *connected* graph, $\text{Jac}(G)$ is a finite abelian group (for a disconnected graph $\text{Div}^0(G)/\text{im } L$ can be infinite, e.g. $\mathbb{Z}$ for two isolated vertices; every graph used below is connected); its “cyclic factors” are the nontrivial cyclic summands in a decomposition of $\text{Jac}(G)$ as a direct sum of cyclic groups. “Zero cyclic factors” (the text’s gloss: “ Jac entirely cyclic-free”) therefore means $\text{Jac}(G) = 0$. The conjecture asserts

$$\text{Jac}(G) = 0 \iff G \text{ is a subdivision of a closed Eulerian graph.}$$

We refute the direction $\Leftarrow$ for the infinite family of all cycles C_m , $m \geq 3$. These include C_3 , the smallest closed Eulerian simple graph with an edge, so they are not “trivial exceptions”.

Not refuted. If “cyclic $\mathbb{Z}$ -components” meant summands isomorphic to the infinite cyclic group $\mathbb{Z}$, then every connected graph would have zero of them, because $\text{Jac}(G)$ is finite. The conjecture would then say that every connected graph is a subdivision of a closed Eulerian graph. That is also false, for example for K_2 , but this reading is not formalized here.

2 Definitions

For a finite vertex set V let $L = D - A$ be the integer Laplacian of G , which is `Mathlib’s SimpleGraph.lapMatrix`.

Definition 1. $\text{Div}^0(G) = \{x \in \mathbb{Z}^V : \sum_v x_v = 0\}$, $\text{Prin}(G) = L\mathbb{Z}^V$ (contained in Div^0 , Lean `prin_le_divZero`), and $\text{Jac}(G) = \text{Div}^0(G)/\text{Prin}(G)$. For a sink $q \in V$ let Δ' be L with the row and column of q deleted. The second standard model is the cokernel $\mathbb{Z}^{V \setminus q}/\Delta'\mathbb{Z}^{V \setminus q}$.

Wikipedia, *Abelian sandpile model* (https://en.wikipedia.org/wiki/Abelian_sandpile_model), calls the second model “the cokernel of the reduced graph Laplacian” and says that it “is most commonly referred to as the sandpile group. Other common names for the same group are critical group, Jacobian group or (less often) Picard group.” We prove the result for both models.

Definition 2. A walk is an *Eulerian circuit* if it is closed and uses every edge exactly once. Wikipedia, *Eulerian path* (https://en.wikipedia.org/wiki/Eulerian_path): “an Eulerian circuit or Eulerian cycle is an Eulerian trail that starts and ends on the same vertex.” A graph is *closed Eulerian* if it is connected and has an Eulerian circuit. Wikipedia mentions a second meaning of “Eulerian graph”: “a graph with every vertex of even degree”. C_m satisfies this one too, since every vertex has degree 2.

Definition 3. Wikipedia, *Homeomorphism (graph theory)*: “The subdivision of some edge e with endpoints $\{u, v\}$ yields a graph containing one new vertex w , and with an edge set replacing e by two new edges, $\{u, w\}$ and $\{w, v\}$.” H is a *subdivision* of G if H is isomorphic to a graph obtained from G by finitely many such steps, possibly none.

3 The counterexample

Let C_m ($m \geq 3$) be the cycle on $\mathbb{Z}/m$, with $i \sim i \pm 1$ (Mathlib’s `cycleGraph m`).

Lemma 4. C_m is closed Eulerian, hence a subdivision (with no subdivided edge) of a closed Eulerian graph.

Proof. C_m is connected. Mathlib’s closed walk `cycleGraph.cycle` is a cycle of length m , so it is a trail with m distinct edges. All m degrees equal 2, so by the handshake lemma C_m has exactly m edges, and the walk uses each of them exactly once. $\square$

Lemma 5. Define $\varphi : \mathbb{Z}^{\mathbb{Z}/m} \rightarrow \mathbb{Z}/m$ by $\varphi(x) = \sum_i i x_i \bmod m$. Then $\varphi(Lf) = 0$ for every f , and $\varphi(\delta_1 - \delta_0) = 1$.

Proof. $(Lf)_i = 2f_i - f_{i-1} - f_{i+1}$. Reindexing, $\varphi(Lf) = \sum_j f_j (2j - (j+1) - (j-1)) = 0$ in $\mathbb{Z}/m$. The second claim is immediate. $\square$

Theorem 6. For every $m \geq 3$, φ induces a surjection $\text{Jac}(C_m) \rightarrow \mathbb{Z}/m$. The restriction $\varphi'(y) = \sum_{i \neq 0} i y_i$ induces a surjection $\mathbb{Z}^{V \setminus 0} / \Delta' \mathbb{Z}^{V \setminus 0} \rightarrow \mathbb{Z}/m$. In particular $\text{Jac}(C_m) \neq 0$, and every decomposition of $\text{Jac}(C_m)$ as a direct sum has a nontrivial summand. Together with Lemma 4, the class “subdivisions of closed Eulerian graphs” contains graphs with a nonzero number of cyclic factors, so the classification is false.

Proof. By Lemma 5, φ kills Prin , and the class of $\delta_1 - \delta_0$ maps to the generator 1. For the reduced model, extend $f \in \mathbb{Z}^{V \setminus 0}$ by 0 at the sink. Then $(\Delta' f)_i = (Lf)_i$ for $i \neq 0$, and the weight of vertex 0 is 0, so $\varphi'(\Delta' f) = \varphi(Lf) = 0$; the class of δ_1 maps to 1. If all summands of a direct sum were trivial, the sum would be trivial, which contradicts $\text{Jac}(C_m) \neq 0$. $\square$

Remark (not formalized). By the same Wikipedia article, “the order of the sandpile group is the determinant of Δ' , which by the matrix tree theorem is the number of spanning trees”. C_m has m spanning trees, so the surjection is an isomorphism: $\text{Jac}(C_m) \cong \mathbb{Z}/m$.

4 Lean formalization

File `lean/Conjecture2154/Basic.lean` (324 lines, only import `Mathlib`), namespace `C2154`.

- `divZero, prin (range of mulVecLin (lapMatrix Z G)), prin_le_divZero, Jac G := divZero V / (prin G).comap (divZero V).subtype.`
- `redLap G q (submatrix of lapMatrix), SandpileGroup G q (its cokernel).`
- `subdivideEdge H u v (new vertex none : Option W), the inductive IsSubdivision (constructors refl, step, iso), IsClosedEulerian (Connected and a closed Walk.IsEulerian), IsSubdivOfClosedEulerian.`
- `cycleGraph_isClosedEulerian, cycleGraph_isSubdivOfClosedEulerian (Lemma 4).`
- `lap_cycle_apply, phi_lap, phi_witness (Lemma 5); jacToZMod_surjective, sandpileToZMod_surjective, jac_cycle_nontrivial, jac_cycle_has_cyclic_factor (Theorem 6).`
- Main: `not_classification : forall n, IsSubdivOfClosedEulerian (cycleGraph (n+3)) /\ Nontrivial (Jac (cycleGraph (n+3))) /\ (surjection Jac -> ZMod (n+3)) /\ (surjection SandpileGroup -> ZMod (n+3)); and not_conjecture.`

Axiom audit: `#print axioms` for `not_classification`, `not_conjecture` and `jac_cycle_has_cyclic_factor` each reports `[proptext, Classical.choice, Quot.sound]`. No `sorry`, no `native_decide`, no added axioms.

5 Relation to the earlier submission

PR #267 (by orionsheep) used C_6 , a subdivision of C_3 , with $\text{Jac}(C_6) \cong \mathbb{Z}/6$. Its Lean theorem proved only $3 \cdot 2 = 6 \wedge 2 \bmod 2 = 0 \wedge 6 = 2 \cdot 3 \wedge 6 > 1 \wedge 2 \neq 0 \wedge 3 \neq 0$. The review rejected it because the decisive content “ $(\text{Jac}(C_6) = \mathbb{Z}/6 \text{ via Matrix-Tree and Smith normal form, and that } C_6 \text{ subdivides the Eulerian } C_3)$ ” is prose- and script-only, with ‘ $6 > 1$ ’ standing in for nontriviality by fiat.” This submission defines the critical group in Lean (both Div^0/Prin and the reduced-Laplacian cokernel, from `Mathlib`’s Laplacian), the subdivision relation, and closed Eulerian graphs. It proves nontriviality through an explicit surjection onto $\mathbb{Z}/m$ for every cycle C_m , $m \geq 3$, not for a single graph. It uses the trivial subdivision (C_m itself), which is already in the class; the isomorphism $\text{Jac}(C_m) \cong \mathbb{Z}/m$ is not needed and is left as a remark.